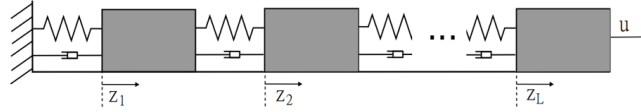


Project 3 - Control of a chain of mass-spring-damper systems

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Abstract

The aim of this practice is to design a stabilizing controller for a chain of $L = 6$ mass-spring-damper systems. The schema of the system under control is reported in the figure below.



Each subsystem has mass $M = 1$, spring constant $k = 10$ and damping constant $h = 2$. The state of the system is given by $x = [z_1, \dot{z}_1, \dots, z_L, \dot{z}_L]'$. The system output is the position of the first mass z_1 , i.e.

$$y(t) = z_1(t)$$

The system can be described by differential equations. The equations in odd position ($i = 2j - 1$, with $j = 1, \dots, 6$), that are referred to the positions of the masses, are

$$\dot{x}_i(t) = x_{i+1}(t)$$

The differential equations for the velocities of the first and the last mass are

$$\dot{x}_2(t) = -\frac{2k}{M}x_1(t) - \frac{2h}{M}x_2(t) + \frac{k}{M}x_3(t) + \frac{h}{M}x_4(t)$$

$$\dot{x}_{12}(t) = \frac{k}{M}x_9(t) + \frac{h}{M}x_{10}(t) - \frac{k}{M}x_{11}(t) - \frac{h}{M}x_{12}(t) + \frac{1}{M}u(t)$$

For the central masses, the equations of the velocities (equations in positions $i = 2j$, with $j = 2, \dots, 5$) are

$$\dot{x}_i(t) = \frac{k}{M}x_{i-3}(t) + \frac{h}{M}x_{i-2}(t) - \frac{2k}{M}x_{i-1}(t) - \frac{2h}{M}x_i(t) + \frac{k}{M}x_{i+1}(t) + \frac{h}{M}x_{i+2}(t)$$

The sampling period of the system is $T_s = 0.1$. The control action u is bounded in $[-15, 15]$, while the positions of the masses are limited in $[-10, 10]$. The velocities are unconstrained. The objective of the control is to drive the system to the equilibrium state corresponding to the output $\bar{y} = \bar{z}_1 = 1$.

To implement the LQR and MPC control, consider as cost matrices

$$Q = \begin{bmatrix} 1+q & 0 & 0 & \dots & 0 \\ 0 & q & 0 & \dots & 0 \\ 0 & 0 & q & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & q \end{bmatrix}, \quad R = r$$

with $q = 10^{-5}$ and $r = 10^{-6}$.

Suggestion: Once you have the MPC code running, you can set in the **quadprog** function the option '**display**' to '**off**'. This will make the simulation a bit faster. Check the help in **quadprog** to see how you pass options.

Exercises

1. Design a stabilizing LQ controller (both in continuous and discrete time) for arbitrary initial conditions chosen by the user (Hint: **dlqr** and **c2d** may be helpful MATLAB commands).
2. Create an MPC controller without any active constraints. How is it possible to reach the LQR performances?
3. Create an MPC controller without any active constraints, considering the terminal cost matrix $S = 0$. Study the closed-loop behavior for different values of the prediction horizon N , considering both small values (e.g. $N = 3$, $N = 6$) and large values ($N \geq 20$). How does the behavior change? How does the prediction horizon affect stability and performances?
4. Add control constraints. Is it still working? Does it perform better than the LQR?
5. Add the constraints on the states, and perform a simulation with initial state $x_0 = [9, 0, \dots, 9, 0]'$. (Hint: in case of infeasibility of the optimization problem, a possible solution is to enlarge the prediction horizon).
6. Suppose that the state is not measurable and that the system is affected by process and measurement noise. Study the effect of the introduction of a Kalman filter in the schema with the MPC on the overall performances.
7. Suppose that there is some uncertainty in the model used by the MPC, and that the real value of the spring constant is $k = 8$. Is the control still working? What happens if the input is penalised more? Implement a velocity form MPC to achieve zero steady state error.

References

- [1] J. Köhler, M. N. Zeilinger, and L. Grüne: *Stability and performance analysis of NMPC: Detectable stage costs and general terminal costs*, IEEE Transactions on Automatic Control, 2023